Electronic Chessboard

«ST-Mate»

USER MANUAL

Version 1.0

ST-Mate

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1 Introduction

This document contains information necessary for the proper operation of the automated system «ST-Mate». The document describes the purpose and operating conditions, preparation for use, description of operations, user actions, as well as recommendations for troubleshooting and guidance on mastering the system.

1.1 Scope of Application

«ST-Mate» is designed as an intelligent chessboard for playing classical chess. It is suitable for use in home environments, educational institutions, chess clubs and other places.

1.2 Brief Description of Automation Capabilities

«ST-Mate» provides the following functions:

- Automatic detection of piece positions on the board.
- Highlighting of legal moves for each piece.
- Move suggestions at the press of a button.
- Game management in three modes.
- Bluetooth wireless interface for remote monitoring and control.

1.3 List of Operational Documentation

Operational documentation includes: User Manual (this document) - contains information for users about the purpose, operating conditions, preparation for use, description of operations, and troubleshooting recommendations.

2 Purpose and Operating Conditions

2.1 Purpose of ST-Mate

«ST-Mate» is an intelligent chessboard designed for playing classical chess in three modes:

- Player controlling white pieces against the built-in chess engine;
- Two-player mode without engine intervention.
- Player controlling black pieces against the built-in chess engine;

«ST-Mate» automatically recognises piece positions, highlights legal moves, provides hints, and manages the game.

2.2 Technical Specifications

- Overall dimensions (L×W×H): 455×455×45 mm.
- Playing field size: 400×400 mm.
- Square size: 50×50 mm.
- Power connector: USB Type-C.
- Supply voltage: 5 V.
- Maximum current consumption: 500 mA.
- Microcontroller: STM32F103C8T6.
- Piece position sensors: 64 Hall effect sensors SS49E.
- Square illumination: addressable LED strip WS2813.
- Chess engine: micro-Max version 4.8 by H.G. Muller.
- Wireless interface: Bluetooth, device name «ST-Mate».

2.3 Operating Conditions

The «ST-Mate» chessboard is intended for operation at temperatures from +5 to +30 °C and relative humidity not exceeding 80 % without condensation. Exposure to direct sunlight, vibration, shocks, or liquids is not allowed.

3 Preparation for Use

3.1 Unpacking and Inspection

Remove the «ST-Mate» chessboard from its packaging. Ensure there is no visible mechanical damage to the housing, board, or pieces. Check the contents.

The package includes:

- Electronic chessboard - 1 pc.
- Pieces with magnets (32 pcs.) - 1 set.
- User manual - 1 pc.

3.2 Connecting to a Power Source

Make sure the power supply (not included) meets the requirements (USB power adapter 5 V / at least 500 mA or a USB port of a computer). The USB Type-C cable (not included) must be undamaged, and the connectors must be free of dirt or deformation.
For first-time use, perform the following steps with no pieces on the board:

1. Connect one end of the cable to the power source.
2. Connect the other end (USB Type-C) to the connector on the rear panel of the ST-Mate.
3. After power is applied, the «ST-Mate» chessboard performs a self-test and calibration: all LEDs briefly light up, then the board enters the waiting state for the initial setup of pieces.

3.3 Functionality Check

After connecting the power and completing the self-test, verify the following:

1. Orient the board so that the nearest corner square on the right side of each player is white.
2. The first two ranks (ranks 1 and 2) from the white side and the two ranks (ranks 7 and 8) from the black side should flash red.
3. Squares on which pieces are placed are illuminated in green.
4. Ranks 3–6 are not lit if they have no pieces; otherwise they flash red. See section 4.3 Initial Position of Pieces.
5. If white and black pieces are placed on the wrong sides (sides swapped), they flash red. See section 4.3 Initial Position of Pieces.

If any of the above conditions are not met, perform a reset using the «RESET» button and repeat the verification procedure. If the problem persists, refer to section 5 Troubleshooting.

4 Description of Operations

4.1 Controls

On the front panel (from left to right):

1. Logo.
2. **«HINT»** button - when pressed, shows a suggested move for the side whose turn it is.
3. Switch - has three positions corresponding to the game modes.
4. **«RNG»** button (random number generator) - activates the mode in which the engine makes random moves (useful for training).

On the rear side of the board (from left to right):

1. **RESET** button - forces a microcontroller reboot.
2. **USB** power connector.

4.2 Selecting the Game Mode

The switch is located on the front panel of the «ST-Mate» chessboard.

Game modes:

- **WvE** - the player controls the white pieces; the controller moves black.
- **PvP** - both players move alternately; the controller only tracks piece positions and provides hints.
- **BvE** - the player controls the black pieces; the controller moves white.

When the mode is changed using the switch, the game is automatically reset. To start a new game, the pieces must be placed in the initial position (see section 4.3 Initial Position of Pieces).

After setting up the pieces and selecting the mode, the «ST-Mate» chessboard is ready for play.

4.3 Initial Position of Pieces

Before starting a game, place the pieces in the initial position corresponding to the standard starting position (Fig. 1). The pieces are recognised automatically by the built-in Hall effect sensors.

Figure 1: Initial position of pieces on the board

Important: After each mode change or when power is applied, the board automatically enters the waiting state for the initial setup. The pieces must be manually placed in their correct squares. Once the setup is complete, the controller automatically detects all pieces and is ready for play.

4.4 Performing a Move

According to chess rules, players alternate moves. White moves first. The move procedure includes the following steps:

1. The player whose turn it is picks up a piece of their colour and places it on the destination square. The Hall effect sensors detect the change in piece positions.
2. If the move is legal according to chess rules, the controller confirms it by highlighting the squares. The move is considered made, and the turn passes to the opponent (or the controller).
3. If an illegal move is attempted (e.g., moving to a square not reachable by the piece, or moving into check), the squares flash red and the move is not accepted. The player must return the piece to its original square and choose another move.

Capturing a piece involves the following steps:

1. The player whose turn it is picks up a piece of their colour.
2. The player also picks up the opponent's piece.
3. The player places their own piece on the square previously occupied by the opponent's piece.

If the capture is legal, the controller confirms the move by highlighting the squares involved.

4.5 Special Moves

Castling: A special move consisting of moving the king two squares horizontally toward a rook of the same colour, and then moving the rook to the square adjacent to the king on the opposite side. Each side may castle once during the game.

Castling kingside is shown in Fig. 2.

Figure 2: Castling kingside

Castling queenside is shown in Fig. 3.

Figure 3: Castling queenside

The castling procedure includes the following steps:

1. The player whose turn it is picks up their king and places it on the destination square.

2. Then they move their rook to the square adjacent to the king on the opposite side.
3. The castling procedure will indicate the positions of the king and rook.
4. If the move is legal, the controller confirms it by highlighting the squares. The turn passes to the opponent (or the controller).
5. If an illegal move is attempted (e.g., moving through check or into check), the squares flash red and the move is not accepted. The player must return the pieces to their original squares and choose another move.

En passant capture: A special pawn capture where a pawn captures an opponent's pawn that has just advanced two squares from its starting square. The capturing pawn moves to the square that the opponent's pawn passed over, as if the opponent's pawn had advanced only one square. The capture is only possible on the move immediately following the opponent's pawn advance.

The en passant capture is shown in Fig. 4.

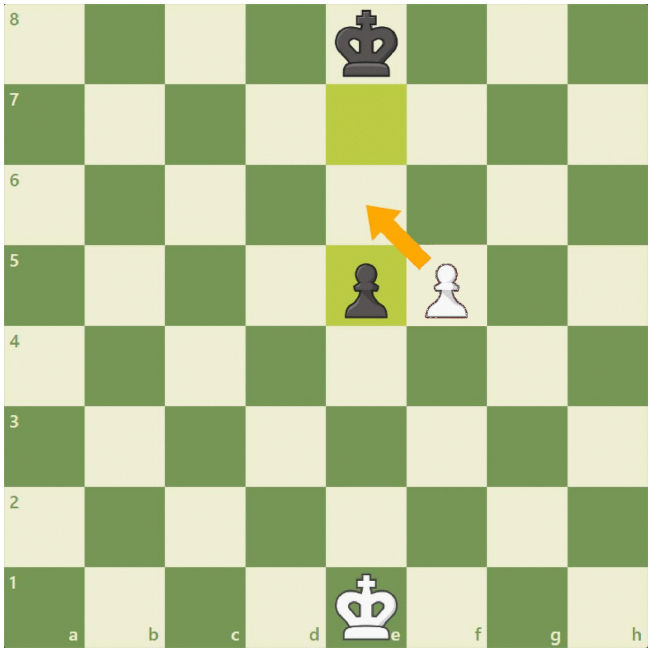

Figure 4: En passant capture

The en passant procedure includes the following steps:

1. The player whose turn it is picks up their pawn.
2. They then pick up the opponent's pawn and remove it from the board.
3. They place their own pawn on the square that was passed over by the opponent's pawn.
4. If the move is legal, the controller confirms it by highlighting the squares. The turn passes to the opponent (or the controller).
5. If an illegal move is attempted, the squares flash red and the move is not accepted. The player must return the piece to its original square and choose another move.

Promotion: A special move in which a pawn that reaches the last rank is promoted to another piece. In the current software version, the pawn automatically promotes to a queen. The option to choose a piece will be added in future updates.

The promotion procedure includes the following steps:

1. The player whose turn it is picks up their pawn on the second last rank.
2. They replace it with a queen of the same colour and place it on the destination square.
3. If a capture is available on the last rank, the player makes the capture and places the queen on that square.
4. If the move is legal, the controller confirms it by highlighting the squares. The turn passes to the opponent (or the controller).
5. If an illegal move is attempted, the squares flash red and the move is not accepted. The player must return the piece to its original square and choose another move.

4.6 Using the HINT Button

The «HINT» button is located on the side panel of the «ST-Mate» chessboard. The hint does not affect the turn order and does not automatically make a move.

4.7 Using the RNG Button

The «PRNG» button is located on the side panel of the «ST-Mate» chessboard. The RNG button is intended to enable random move generation by the controller in modes involving the chess engine. The RNG mode can only be activated when the board is powered on or reset.

4.8 Reset and Restart

If a forced reset of the «ST-Mate» chessboard is required (e.g., in case of a freeze or to start a new game), use the «RESET» button located next to the USB Type-C connector. A short press of the button reboots the microcontroller and repeats the initialisation procedure. After a reset, the pieces must be placed in the initial position again.

4.9 Bluetooth Control

For remote control and monitoring, the board is equipped with a Bluetooth interface. Connect to it using a smartphone, tablet, or computer with any terminal client (e.g., *Serial Bluetooth Terminal*). After connection, a text-based command protocol is available. Commands are transmitted in UTF-8 encoding and end with the characters `\r\n` (carriage return and line feed).

4.10 Connection

When the «ST-Mate» chessboard is powered on, it becomes available for Bluetooth connection under the name `ST-Mate`.

5 Troubleshooting

Table 1 lists possible abnormal situations and recommended actions.

Situation	User Action
The «ST-Mate» chessboard does not turn on, no backlight	Check the USB Type-C cable connection, ensure the power supply provides 5 V and at least 500 mA. Try a different cable or USB port.
Pieces are not recognised (no reaction to movement)	Ensure the pieces are from the original set and placed exactly in the centre of squares. If necessary, press «RESET» and set up the pieces again.
Erratic backlight (random flashing)	Possible influence of external magnetic fields. Remove sources of strong magnetic fields. Perform a reset.
Piece highlighting does not match their positions	Increase the sensor trigger level in settings by 5–10 units. Increase the smoothing coefficient in settings by 1–2 units. Check the power supply for compliance with requirements. Reduce backlight brightness in settings.
After calibration, sensors do not react to pieces	Make sure calibration was performed with no pieces on the board. Repeat forced calibration. Adjust the trigger level and smoothing coefficient.
The engine does not make a move in modes with controller	Verify the correct mode is selected and that it is the controller's turn. If the delay exceeds 10 seconds, press «HINT» to initiate a move.
Bluetooth does not work, ST-Mate not in device list	Ensure ST-Mate is powered on. Contact the ST-Mate developer.

Table 1: Abnormal situations and remedies

6 Guidance for Mastering

To quickly become familiar with the «ST-Mate» chessboard, it is recommended to:

- Carefully read this manual, especially sections 4 Description of Operations and 5 Troubleshooting.
- Familiarise yourself with the FIDE laws of chess. Link to the official FIDE handbook: https://handbook.fide.com/ (accessed: 01.04.2026).
- Start with «Player vs Player» mode without using hints, to get used to the physical interaction with the board and understand how moves are recognised.
- After mastering basic operations, move on to playing against the controller, using hints to study tactical techniques.
- If in doubt about the correctness of the piece positions, use the «Reset» button and start a new game.

6.1 Visual User Messages

The «ST-Mate» chessboard uses various methods to inform the user about the current state and events. It is advisable to use the standard colour theme (THEME:0) for mastering, which intuitively displays legal moves and errors. The following examples of light indication are given for the standard theme.

For the initial setup and waiting for the game to start, squares of ranks 1 and 2 from the white side and ranks 7 and 8 from the black side are lit in red, while the four central ranks are

not lit. As soon as the pieces are placed in their correct squares, the board recognises them and turns off the backlight. When a piece is selected, the square it occupies starts flashing white. Legal moves for the selected piece are illuminated in white. Once the piece is moved to a new square, the completed move is highlighted in yellow, and the turn passes to the opponent (or the controller). If an illegal move is attempted, the squares flash red. When the «HINT» button is pressed to show a recommended move, the squares involved in the suggestion are lit in blue. During castling, the squares where the king moves are lit white. After the king moves, the completed move is highlighted yellow. Then the rook move is indicated with white light; after the rook is moved, the turn passes to the opponent (or the controller). When delivering a check, the attacking piece flashes blue, while the king in check flashes red. When checkmate occurs, the attacking piece flashes green, and the mated king flashes red. The game ends. In a stalemate situation, the kings flash yellow. The game ends. After the game ends, any movement of a piece resets the board to waiting for the initial setup for a new game, and ranks 1–2 from white and 7–8 from black light up red again.

6.2 Text User Messages via Bluetooth

Using Bluetooth to obtain detailed information about moves and game state can be useful for analysis and learning, but is not required for basic operation. Via Bluetooth, ST-Mate can send text messages informing the user about the current state, move results, and other events. Completed moves are output in algebraic notation (e.g., "e4" for a pawn move from e2 to e4). Moves that were rejected due to rule violations are accompanied by an error message (e.g., "White make invalid move e2e5"). At the end of the game, the result is displayed (e.g., "1-0" for white win, "0-1" for black win, "1/2-1/2" for draw). The PGN of the current game is also output, allowing it to be saved and analysed. Output for initialisation, start, game progress, and result in PGN format is shown in Fig. 5.

```
Voltage CPU: 3.274: OK
Flash state: OK
DateTime: 13.04.2026 22:56:02
Mode: PvP
RNG: ON. seed: 397067
Initialization...successes
##################################################
#     ____________       __   ____          __   #
#    / ____/_  __/      /  |/  /____    / /_ ___ #
#    \__ \ / /_______ / /|_/ //__  \ / __// _ \ #
#   ___/ // //_____// /  / // _   // /_ /  __/ #
# /____//_/       /_/  /_/ \__,_/ \__/ \___/   #
#                                                #
##################################################
 ST-Mate v1.0.0 (c)
 Developed by: Alexander Kondratyev
 System:  STM32F103C8 | 64kB Flash | 20kB RAM
 Engine:  micro-Max v4.8 by H.G. Muller
##################################################

New game start
1. e4
1. e5
2. Bf1c4
2. Bf8c5
3. Qd1f3
3. Nb8c6
4. Qf3xf7#

[Event "Casual Game"]
[Site "Iserlohn, NRW GER"]
[Date "2026.04.13"]
[Round "1"]
[White "player1"]
[Black "player2"]
[Result "1-0"]

1. e4 e5 2. Bf1c4 Bf8c5 3. Qd1f3 Nb8c6
4. Qf3xf7# 1-0
```

Figure 5: Bluetooth message log

6.3 Sending Commands

After connecting to the board via Bluetooth, the user can send text commands. Commands are case-insensitive. Supported queries and commands are listed in Appendix A. Command Protocol, and also when sending the command HELP\r\n, a help message with all available commands is displayed.

6.4 Usage Examples

```
> BRIGHT:80\r\n
Brightness set:80\r\n
> THEME:5\r\n
Theme set:5\r\n
> TRIGLV?\r\n
Trigger level:40\r\n
```

7 Technical Maintenance

To avoid injury and damage, as well as to ensure a long service life of the «ST-Mate» chessboard, the following maintenance recommendations should be observed.

- Do not disassemble the device yourself, as this may cause injury or damage.
- Keep the board and pieces clean; avoid liquid ingress into the housing.
- Regularly check the integrity of the power cable and the USB Type-C connector. Do not use damaged cables.
- For storage and transportation, use the original packaging or equivalent protection against shocks and vibrations.
- Do not expose the device to extreme temperatures, direct sunlight, or strong magnetic fields.

If malfunctions occur, refer to the "Troubleshooting" section or contact the developers.

7.1 Sensor Calibration

Calibration is recommended upon first use, after replacing the power supply or cable, or when operating conditions (temperature, humidity) change. The calibration table is stored in non-volatile memory. Calibration is performed automatically at power-on or reset if there are no pieces on the board.
There are two ways to perform forced calibration:

1. With the «HINT» button held down when applying power or pressing the «RESET» button.
2. Via the terminal command `CALIBRATE\r\n` over Bluetooth.

To perform forced calibration:

- Remove all pieces from the board.
- Use one of the methods to start calibration.
- Wait for the process to finish (about 5 seconds).

7.2 Storage and Transportation

Store the «ST-Mate» chessboard in a dry, clean place at a temperature between +5 and +30 °C. For transportation, use the original packaging or equivalent protection against shocks and vibrations. Avoid exposure to direct sunlight, moisture, extreme temperatures, and mechanical damage.

7.3 Surface Care

Wipe the housing and pieces with a soft, slightly damp cloth. Do not allow liquid to enter the housing. Do not use abrasive cleaning agents.

8 List of Abbreviations

- AS - automated system.
- USB - Universal Serial Bus.
- LED - Light Emitting Diode.
- Bluetooth - wireless technology for short-range data exchange.
- PGN - Portable Game Notation, a format for recording chess games.
- FEN - Forsyth-Edwards Notation, a format for describing a chess position.
- RNG - Random Number Generator.
- STM32 - family of 32-bit microcontrollers from STMicroelectronics.
- SS49E - Hall effect sensor used to detect the presence of a magnetic field.
- WS2813 - type of addressable LED strip.
- micro-Max - compact chess engine developed by H.G. Muller. Source code available at: https://home.hccnet.nl/h.g.muller/umax4_8.c (accessed: 01.04.2026).

9 Developer Information and Contact Details

- **Developer of the ST-Mate chessboard:** Kondratyev Alexander Nikolaevich.
  Email: alexander.kondratyev.dev@gmail.com
  Telegram: https://t.me/kondratyev_an
- **Developer of the micro-Max chess engine:** H.G. Muller.
  Homepage: https://home.hccnet.nl/h.g.muller/ (accessed: 01.04.2026).

All questions regarding operation, configuration, and modification of the «ST-Mate» chessboard should be directed to the developers.

Appendix A. Command Protocol

Command	Description
Service Commands	
`REBOOT`	Reboot the device (analogous to pressing the «RESET» button).
`SERVICE:X`	Enable/disable service mode. X=0 - disabled, X=1 - enabled.
`VOLT?`	Request processor supply voltage (returns value in millivolts).
`CALIBRATE`	Start sensor calibration (board must be empty).
`TRIGLV?`	Request current trigger level of sensors (default 25).
`TRIGLV:X`	Set sensor trigger level (X is an integer depending on calibration).
`ALPHA?`	Request current exponential smoothing coefficient (default 8).
`ALPHA:X`	Set exponential smoothing coefficient (X is an integer).
User Commands	
`HELP`	Request list of available commands.
`EVENT?`	Request tournament or match name.
`EVENT:X`	Set tournament or match name (e.g., EVENT:Casual Game).
`SITE?`	Request venue.
`SITE:X`	Set venue (e.g., SITE:Moscow).
`TIME?`	Request current time.
`TIME:X`	Set current time (X is a string in the format dd.mm.yy HH:MM:SS).
`ROUND?`	Request round number.
`ROUND:X`	Set round number (e.g., ROUND:1).
`WHITE?`	Request name of the player playing white.
`WHITE:X`	Set name of the player playing white (e.g., WHITE:Alice).
`BLACK?`	Request name of the player playing black.
`BLACK:X`	Set name of the player playing black (e.g., BLACK:Bob).
`FEN?`	Request current position in FEN format.
`PGN?`	Request current game in PGN format.
`SEED?`	Request current seed for the random number generator.
`SEED:X`	Set seed for the random number generator (X is an integer or string).
`BRIGHT?`	Request current backlight brightness level.
`BRIGHT:X`	Set backlight brightness (0–100). Example:`BRIGHT:50`.
`THEME:X`	Set colour theme (0–11). List of themes is given in Appendix B. Colour Schemes for Backlight.
`HINTBTN:X`	Enable/disable the HINT button. X=0 - button inactive, X=1 - active.

Appendix B. Colour Schemes for Backlight

Code	Theme Name
0	Default - standard theme (default)
1	Neon Night - bright neon colours
2	Forest Glade - natural green and brown shades
3	Ocean Breeze - marine theme with blue tones
4	Royal Purple - noble purple shades
5	Autumn Sunset - warm autumn colours
6	Arctic Ice - cool pastel tones
7	Desert Sand - sandy and earthy shades
8	Candy Land - bright candy colours
9	Midnight Oil - dark saturated tones
10	Spring Meadow - fresh spring colours
11	Classic Elegance - restrained classic palette